Supplementary Material for As-Rigid-As-Possible Deformation of Gaussian Radiance Fields

1 Additional Results

In this section, we present additional experimental results, including challenging deformation cases and examples on real-world datasets with backgrounds. These experiments illustrate the general applicability of our algorithm.

Fig. 1 illustrates the performance of our algorithm on complex deformations and real-world datasets with backgrounds. The baseline represents the results obtained by applying only the geometric deformation without optimization. We can see our algorithm achieves favorable results in challenging composite deformations involving twisting and bending, as well as on real-world datasets with backgrounds. Fig. 2 presents the multi-view rendering results before and after deformation on real-world datasets.

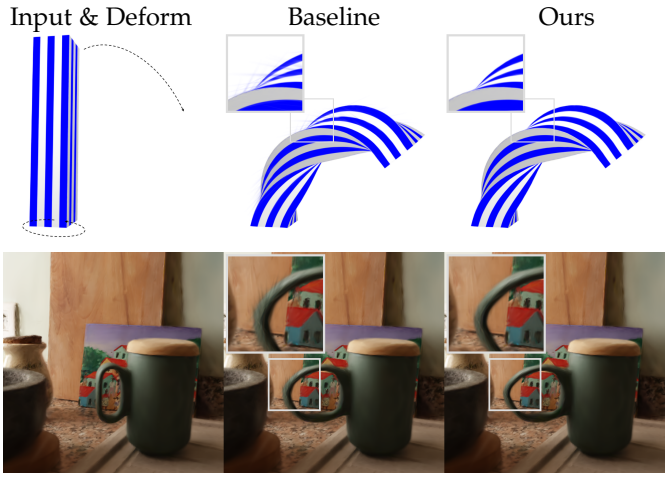


Fig. 1: **Comparisons with baseline on a composite deformation example (twisting and bending) and a real-captured example.** The dataset in the second row of the image is sourced from NeRFshop [1].

2 Hyperparameter Selection and Sensitivity Analysis

In this section, we present the selection of hyperparameters, such as the learning rate and energy term weights, as well as their sensitivity to the algorithm. Overall, the choice of hyperparameters within a reasonable range does not significantly affect the results, demonstrating the stability of our algorithm.

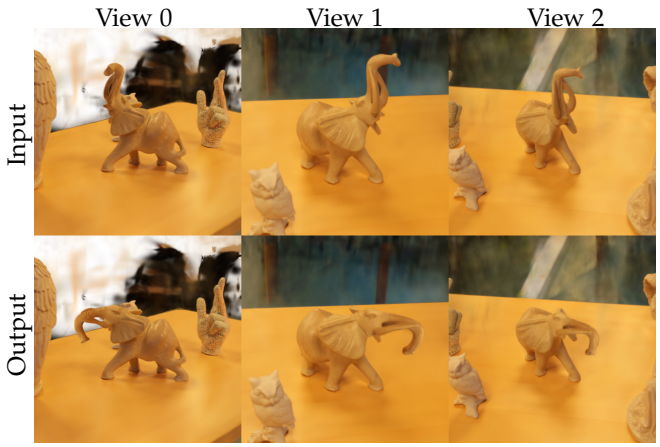


Fig. 2: **Multi-view visualization of deformation results on real-world datasets.** The dataset in the image is sourced from NeRFshop [1].

TABLE 1: **Quantitative analysis of the sensitivity of results to the learning rate.** The results in the table are obtained after optimizing for 100 steps under different learning rate settings.

Dataset	Metrics	0.5× lr	lr	2.0× lr	lr_{3DGS}
Hotdog	PSNR ↑	28.00	**28.02**	**28.02**	27.99
	SSIM ↑	0.9641	**0.9643**	**0.9643**	0.9639
	LPIPS ↓	0.0327	**0.0326**	0.0327	0.0327
Ship	PSNR ↑	**29.60**	**29.60**	29.54	29.58
	SSIM ↑	**0.9321**	**0.9321**	0.9317	0.9319
	LPIPS ↓	**0.0578**	**0.0578**	0.0581	0.0579
Stripes	PSNR ↑	32.78	32.89	**32.93**	32.92
	SSIM ↑	0.9919	0.9924	**0.9927**	0.9921
	LPIPS ↓	0.0053	**0.0051**	**0.0051**	0.0052
Bunny	PSNR ↑	33.78	**33.82**	33.81	33.80
	SSIM ↑	0.9753	0.9756	**0.9758**	0.9753
	LPIPS ↓	0.0315	0.0310	**0.0306**	0.0313
Pinocchio	PSNR ↑	41.45	41.67	**41.77**	41.41
	SSIM ↑	0.9868	**0.9869**	**0.9869**	0.9865
	LPIPS ↓	0.0355	**0.0354**	**0.0354**	0.0358
Glove	PSNR ↑	28.24	28.24	**28.25**	28.24
	SSIM ↑	0.9790	0.9791	**0.9792**	0.9790
	LPIPS ↓	0.0308	0.0305	**0.0304**	0.0306

TABLE 2: **Quantitative experiments on the selection of λ_1.** In this experimental setting, λ_2 is fixed at 99. The term 'average' in the table refers to the mean computed over the cases presented in the six synthetic datasets (*stripes*, *glove*, *pinocchio*, *bunny*, *ship* and *hotdog*), which are reconstructed using the 3DGS method.

Dataset	Metrics	$\lambda_1 = 0.01$	$\lambda_1 = 0.1$	$\lambda_1 = 1$	$\lambda_1 = 10$	$\lambda_1 = 100$
Average	PSNR ↑	32.36	32.36	**32.37**	32.36	32.25
	SSIM ↑	0.9716	**0.9717**	**0.9717**	0.9716	0.9710
	LPIPS ↓	**0.0321**	**0.0321**	**0.0321**	0.0322	0.0328

TABLE 3: **Quantitative experiments on the selection of λ_2.** In this experimental setting, λ_1 is fixed at 1.

Dataset	Metrics	$\lambda_2 = 1$	$\lambda_2 = 10$	$\lambda_2 = 99$	$\lambda_2 = 500$	$\lambda_2 = 1000$
Average	PSNR ↑	32.37	**32.38**	32.37	32.30	32.26
	SSIM ↑	0.9717	**0.9718**	0.9717	0.9715	0.9714
	LPIPS ↓	0.0323	**0.0321**	**0.0321**	0.0322	0.0323

TABLE 4: **The computational cost of establishing the sampled representation of the radiance fields and building the graph for deforming.**

Dataset	Stripes	Pinocchio	Glove	Bunny	Ship	Hotdog	Cup	Shoe
Pre. Time	0.73s	2.14s	1.14s	3.73s	4.58s	2.08s	2.02s	1.58s

2.1 Learning Rates

In this subsection, we present the quantitative results on the impact of the learning rates on the final outcomes. The term lr refers to the learning rates specified in the main text, which are the same across all the experiments in this paper. The term lr_{3DGS} refers to the learning rates used in 3DGS [2] (1×10^{-3} for rotation, 5×10^{-2} for scaling, 2.5×10^{-2} for opacity, 2.5×10^{-3} for 0-th order spherical harmonics, 1.25×10^{-4} for the rest spherical harmonics, the learning rate for the position is exponentially interpolated between $1.6\, r_{scene} \times 10^{-4}$ and $1.6\, r_{scene} \times 10^{-6}$ based on the training steps, r_{scene} is the scaling factor used to normalize the scene) . To evaluate sensitivity, we halve and double the learning rates and repeat the quantitative experiments on all synthetic datasets, as shown in Table. 1. We also report the results using the learning rates from the original 3DGS. The experiments show that our default learning rates, the halved ones, the doubled ones, and the original learning rates in 3DGS all produce comparable results. This indicates that our algorithm is robust to the learning rates within a reasonable range.

2.2 Energy Term Weights

In this subsection, we present the quantitative impact of different energy term weights on our results, as shown in Table. 2 and Table. 3. In Fig. 3, we present visual results on the real dataset for different values of λ_2. The results demonstrate that our results remain comparable within a reasonable range of parameter values. Smaller values of λ_2 may lead to artifacts at the boundaries, we hence adopt the weights of $\lambda_1 = 1, \lambda_2 = 99$ in all the experiments.

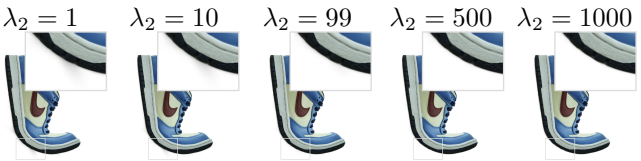


Fig. 3: **Qualitative analysis of λ_2.** In this experimental setting, λ_1 is fixed at 1.

3 Time Required for Preprocessing

In this section, we present the time required to construct the sampled representation of the radiance field and graph structure for as-rigid-as-possible deformation based on a given 3DGS model, as shown in Table 4 (as Pre. Time). This computational cost constitutes a one-time preprocessing overhead and does not affect the frame rate during deformation or the optimization time after geometric deformation. The reported construction time includes building the graph structure, sampling the radiance field, performing forward computation to obtain feature values at sample points, and computing the weights of the K-nearest control points for both Gaussian endpoints and sample points. As shown in the table, the preprocessing time across most datasets ranges from 1 to 3 seconds (on a single RTX 4090).

References

[1] C. Jambon, B. Kerbl, G. Kopanas, S. Diolatzis, T. Leimkühler, and G. Drettakis, "Nerfshop: Interactive editing of neural radiance fields," *Proc. ACM Comput. Graph. Interact. Tech.*, vol. 6, no. 1, pp. 1:1–1:21, 2023.

[2] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, "3d gaussian splatting for real-time radiance field rendering," *ACM Trans. Graph.*, vol. 42, no. 4, pp. 139:1–139:14, 2023.